Pyrolysis Advisor

Using physics + machine learning to get more bio-oil with less oxygen

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THE PROBLEM

Bio-oil contains too much oxygen

Pyrolysis turns biomass into oil, but that oil is 31–52% oxygen — acidic, unstable and hard to use.

Biomass

wood, husk, residue

→

Pyrolysis

heat to 400–600 °C
without air

→

Bio-oil

plus char and gas

Our question: Which temperature and heating rate give the most oil with the least oxygen, for a given biomass?

OUR APPROACH

Three steps: data → models → advice

1

Data

2 published datasets

~1,270 pyrolysis experiments from the literature (Zhang 2022, Pascarella 2025)

2

Physics + ML

Predict oil yield and oxygen (O/C)

Reaction-kinetics features feed XGBoost and Gaussian-process models

3

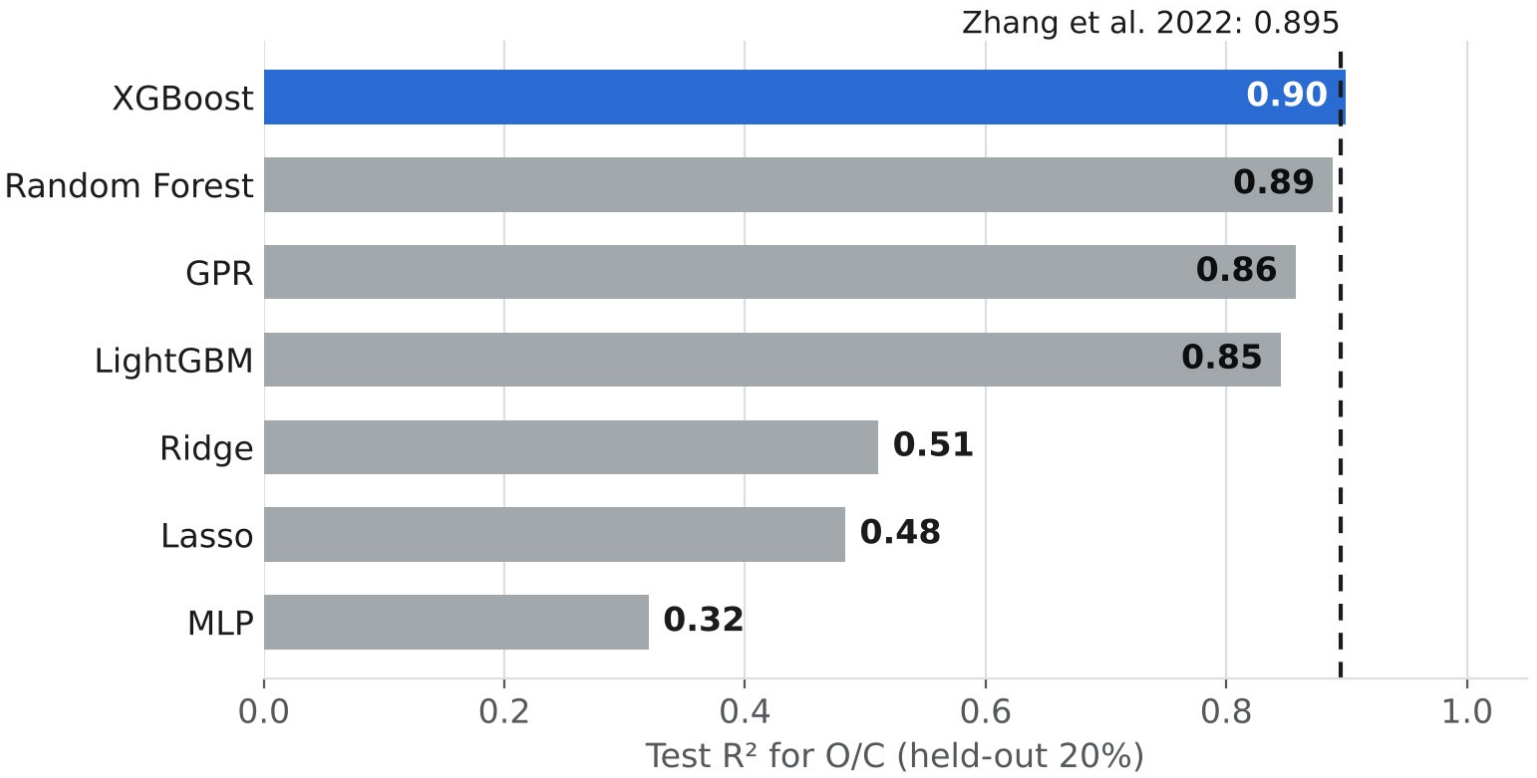
Pyrolysis Advisor

Recommends conditions

Suggests temperature and heating rate, and shows the yield vs oxygen trade-off

RESULT 1 · ACCURACY

As accurate as the published study



R² 0.90

Our best model (XGBoost) on unseen test data

0.895

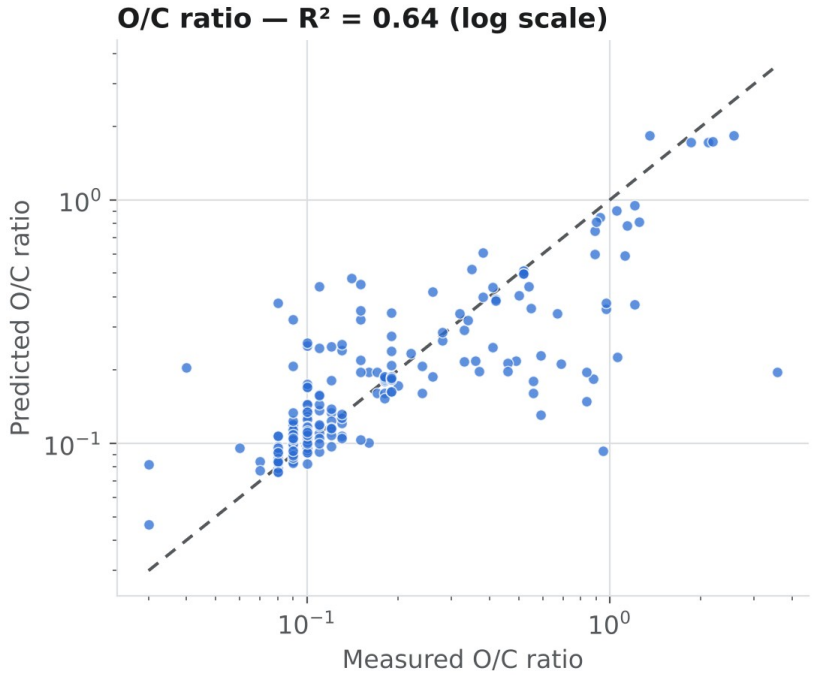
Published result (Zhang et al. 2022)

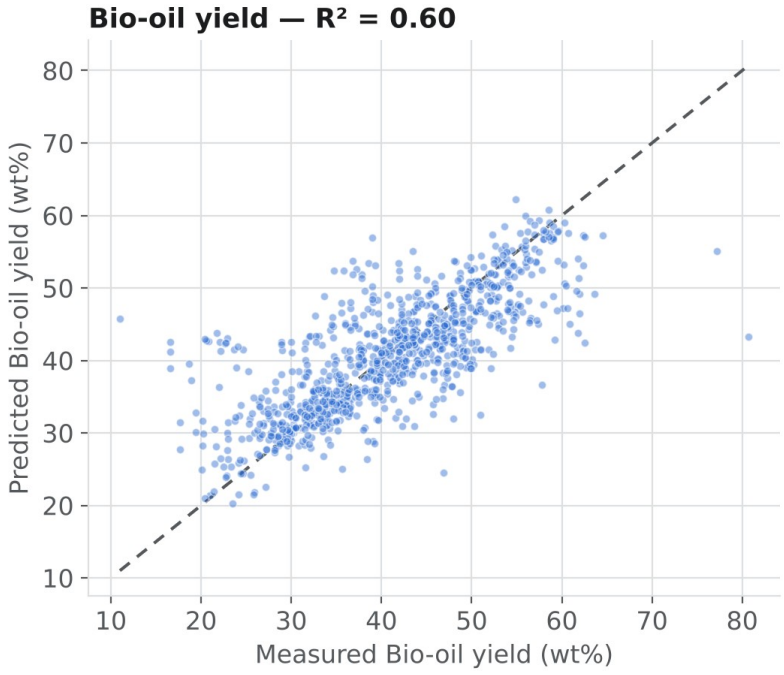
R² = 1 means perfect prediction.

RESULT 2 · PREDICTIONS VS REALITY

Predictions follow the measured values

Each dot is one experiment. Dots on the dashed line = perfect prediction.





± 4 wt%

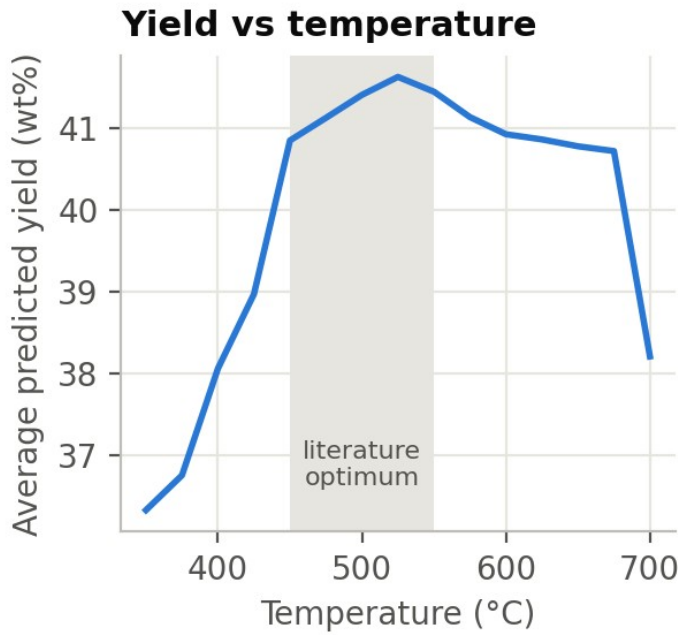
Typical error in predicted oil yield

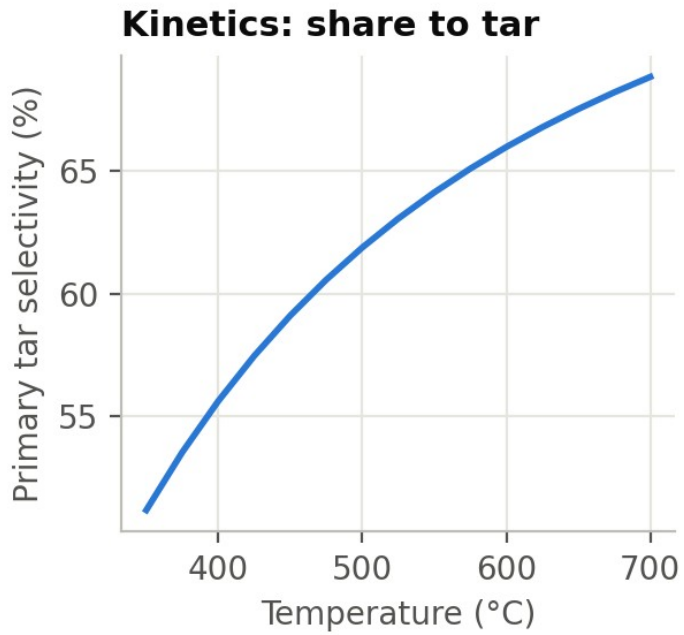
18%

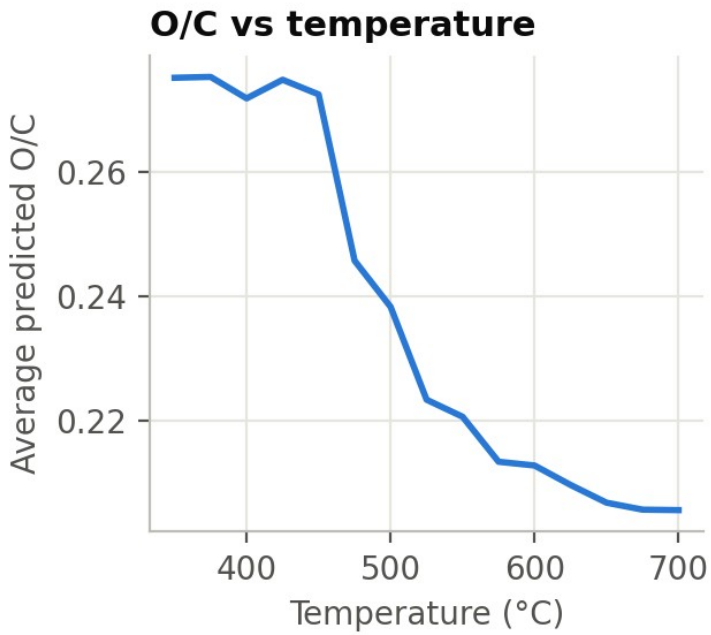
Typical error in predicted oxygen (O/C)

RESULT 3 · PHYSICS CHECK

The model behaves like real pyrolysis chemistry







Oil peaks at ~525 °C

matches the known best range (450–550 °C)

Too hot → less oil

oil vapours crack into gas above ~600 °C

Hotter → less oxygen

oxygen content always falls with temperature

RESULT 4 · THE ADVISOR

It recommends the best reactor settings

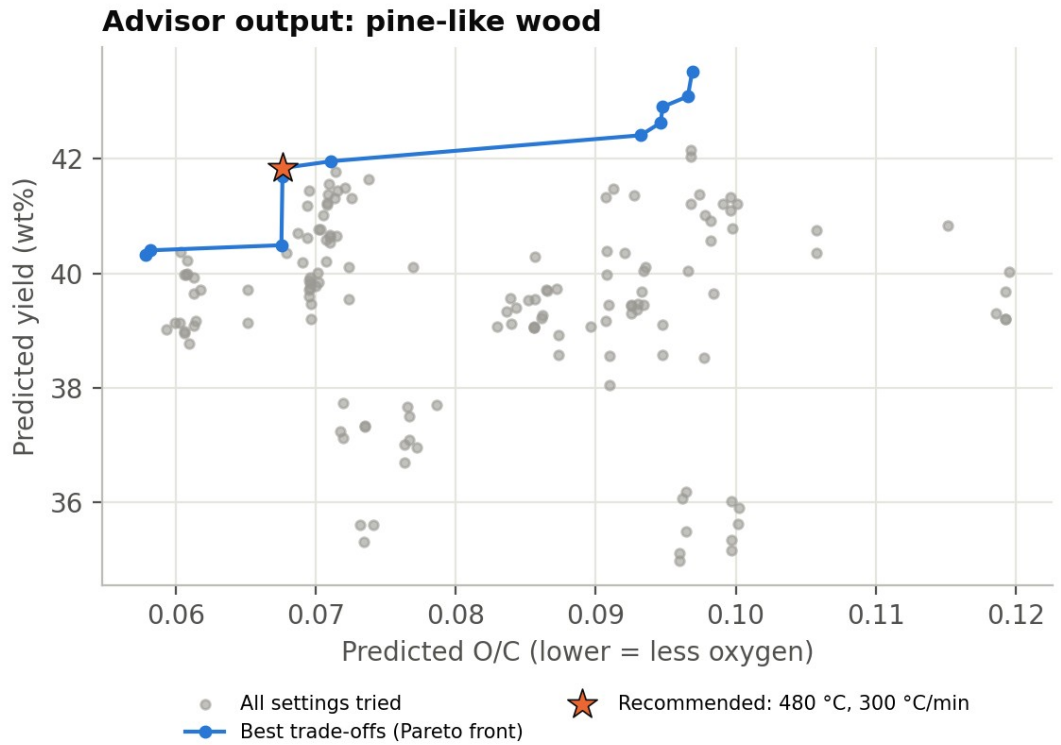
Example: pine wood

RECOMMENDED

480 °C · 300 °C/min

41.8 wt% bio-oil
O/C 0.068 (low oxygen)

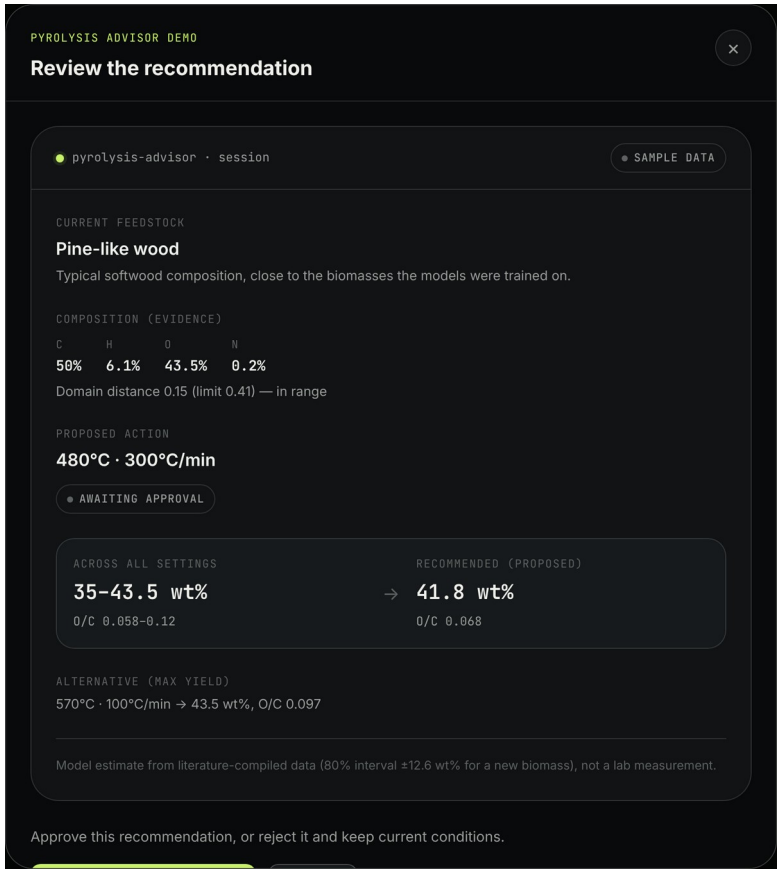
Max-yield option: 570 °C → 43.5 wt% oil, but O/C 0.097 (≈40% more oxygen)

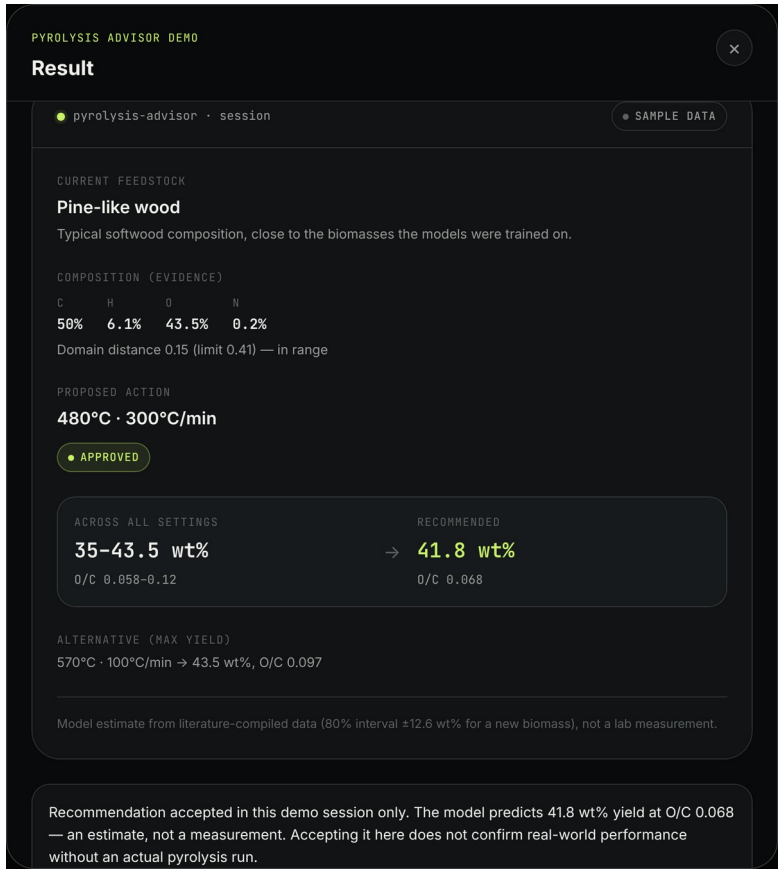


Each dot is one setting the Advisor tested; the line is the best trade-off.

WORKING MODEL

Try it: the Advisor web app



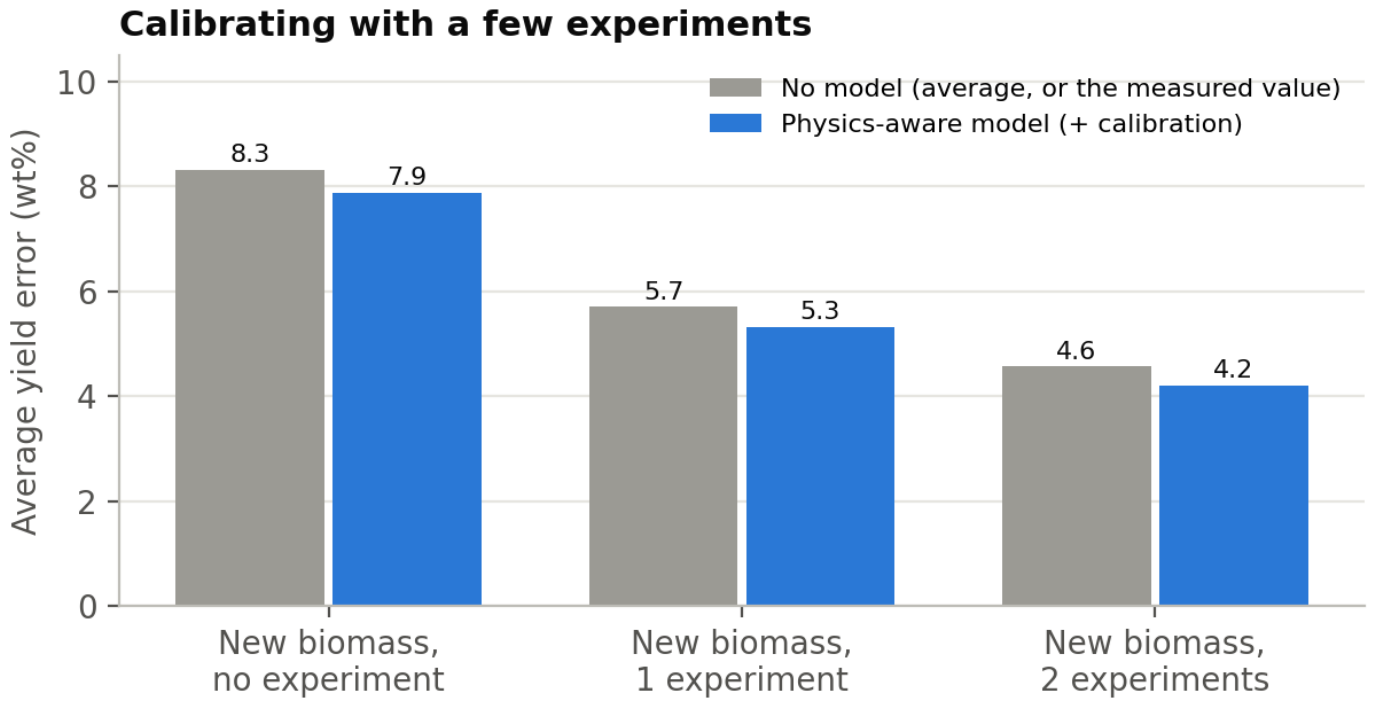


1. Pick a biomass
2. See the recommended temperature and heating rate
3. Accept or reject — the result is shown

RESULT 5 · NEW BIOMASS

New biomass? One or two experiments improve it

Error in predicted oil yield for a biomass the model has never seen



8 → 5.3 → 4.2

Yield error (wt%) with 0, 1 and 2 real experiments

- Brand-new biomass is the hardest case
- The Advisor warns when a biomass is unlike its training data

SUMMARY

What we achieved

R^2 0.90	Oxygen prediction as good as the published study
± 4 wt%	Typical error in predicted oil yield
✓ Physics	Model agrees with known pyrolysis chemistry
Web app	Advisor recommends temperature and heating rate

Thank you — questions?

github.com/umasuthanpalaniappan/BioOil-Automation